# Inventor Mobility, Knowledge Diffusion, and Growth

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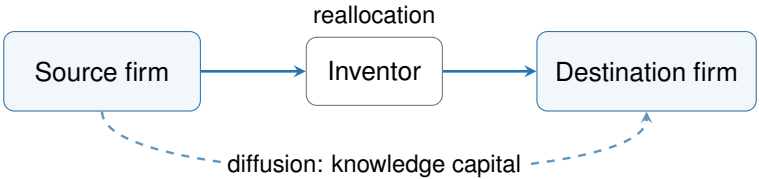
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# How does inventor mobility affect growth and welfare?

- "Mobility of personnel among firms provides a way of spreading information" (Arrow, 1962)




- Non-compete clauses affect inventor mobility, as in Silicon Valley against Route 128 (Saxenian, 1994; Gilson, 1999)
- German inventors' annual job-to-job rate fell from 12.5% in 2001 to 4.8% in 2014

\+ Inventors move to firms that value them more (reallocation)

\+ The destination firm gains part of the source firm's knowledge (diffusion)

– Search is costly, and search and innovation externalities distort mobility

# Motivating Facts and the Model

- German employer–employee matched data linked to patent data
- Patent-listed inventors often move to lower-mean-wage establishments
- Inventors move down more often than other workers yet receive larger wage gains
- A firm-wage job ladder, in which moves go to higher-wage employers, generates neither fact
- An endogenous growth model with on-the-job search, in which a move transfers knowledge
- That knowledge makes moves to less productive firms profitable

# Main Results

- A non-compete ban raises inventor mobility and the annual growth rate
- The 2001–2014 inventor mobility decline lowers the annual growth rate by 0.25 pp
- Knowledge diffusion, not reallocation, drives most of both growth effects
- A non-compete ban lowers consumption-equivalent (CE) welfare by 0.41%

# Literature

- Knowledge diffusion & growth
  - Kortum (1997); Lucas and Moll (2014); Perla and Tonetti (2014); Akcigit et al. (2018); Buera and Oberfield (2020); Benhabib et al. (2021); Prato (2025); Crews (2023); Babalievsky (2023)
  - Job-to-job transition $\rightarrow$ knowledge diffusion
- Inventor mobility & knowledge diffusion
  - Almeida and Kogut (1999), Song et al. (2003), Rosenkopf and Almeida (2003), Hoisl (2007), Breschi and Lissoni (2009), Singh and Agrawal (2011), Kaiser et al. (2015), Rahko (2017), and Braunerhjelm et al. (2020)
  - Quantifying growth effects and policy counterfactuals
- Non-compete clauses
  - Marx et al. (2009); Balasubramanian et al. (2022); Shi (2023); Jeffers (2024); Baslandze and Vardishvili (2026); Gottfries and Jarosch (2026)
  - Moves between incumbent firms that transfer knowledge

# Motivating Facts

# German Inventor and Worker Data

- **INV-BIO.** 152,350 inventors named on EPO patent applications 1999–2011, with their social-security employment and wage histories 1980–2014
- **SIAB.** A 2% random sample of the same records (1,940,962 individuals), without patents
- The two data sets cannot be linked

# Fact 1-1: Many inventor moves go to lower-wage establishments

Inventor moves by establishment mean-wage bin (row %)

| | Destination rank bin | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Origin | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 1 | 18 | 16 | 13 | 12 | 10 | 9 | 7 | 6 | 5 | 4 |
| 2 | 11 | 16 | 14 | 12 | 10 | 9 | 8 | 8 | 7 | 6 |
| 3 | 7 | 13 | 14 | 12 | 10 | 9 | 9 | 9 | 8 | 8 |
| 4 | 6 | 10 | 11 | 12 | 11 | 11 | 11 | 10 | 10 | 8 |
| 5 | 5 | 8 | 9 | 11 | 11 | 11 | 10 | 12 | 11 | 12 |
| 6 | 4 | 7 | 9 | 9 | 10 | 12 | 13 | 12 | 12 | 11 |
| 7 | 4 | 6 | 7 | 8 | 9 | 11 | 13 | 13 | 14 | 15 |
| 8 | 3 | 5 | 6 | 7 | 10 | 11 | 14 | 13 | 14 | 17 |
| 9 | 2 | 5 | 5 | 6 | 9 | 9 | 14 | 14 | 16 | 21 |
| 10 | 2 | 3 | 4 | 4 | 6 | 7 | 10 | 12 | 19 | 32 |

- Ranked by mean wage, 36% of inventor moves go to a lower bin

INV-BIO, 1980–2014, inventor job-to-job moves. Establishments ranked within year by previous-year mean wage, in ten bins formed by cumulative inventor counts.

Moves with wage gains

# Fact 1-2: Many inventor moves go to less productive establishments

Inventor moves by establishment citation-productivity bin (row %)

| | Destination rank bin | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Origin | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 1 | 12 | 12 | 11 | 10 | 8 | 9 | 9 | 9 | 10 | 10 |
| 2 | 12 | 12 | 12 | 9 | 9 | 9 | 9 | 9 | 10 | 10 |
| 3 | 10 | 12 | 12 | 11 | 9 | 10 | 9 | 9 | 9 | 9 |
| 4 | 11 | 10 | 11 | 11 | 11 | 10 | 11 | 9 | 9 | 8 |
| 5 | 9 | 9 | 10 | 9 | 10 | 11 | 15 | 10 | 9 | 10 |
| 6 | 8 | 8 | 10 | 7 | 12 | 9 | 9 | 12 | 11 | 13 |
| 7 | 6 | 8 | 6 | 14 | 12 | 9 | 12 | 11 | 11 | 12 |
| 8 | 5 | 6 | 7 | 7 | 8 | 12 | 12 | 14 | 14 | 14 |
| 9 | 4 | 5 | 10 | 6 | 12 | 8 | 9 | 16 | 13 | 17 |
| 10 | 5 | 7 | 6 | 6 | 6 | 10 | 10 | 11 | 15 | 25 |

- Ranked by citation productivity, 37% of inventor moves go to a lower bin

INV-BIO, 1980–2014, inventor job-to-job moves. Establishments ranked within year by the three-year backward average of citation productivity per inventor (positive values only), in ten bins formed by cumulative inventor counts.

# Fact 2: Inventor-proxy movers move down more often yet earn more

| | Move to lower-mean-wage establishment | Own-wage growth |
|---|---|---|
| Inventor proxy $I_{i,t-1}$ | $0.0080^{***}$ | $0.0581^{***}$ |
| | (0.0021) | (0.0033) |

SIAB job-to-job movers, $N = 651{,}554$. Standard errors two-way clustered by origin and destination establishment.

- Inventor proxies are workers with a university-entrance qualification in the five occupation groups most common among inventors
- Controls: age, age squared, gender, log wage; origin, destination, and year FE
- A firm-wage job ladder, in which moves go to higher-wage employers, generates neither fact

# Model

# Heterogeneous Firms with Knowledge Capital and Inventors

- Continuum of firms with
  1. continuum mass of knowledge capital (or product lines) $k$
  2. continuum mass of inventors $n$
- Fixed total mass of inventors $N$
- Instantaneous payoff

$$\pi k - c\left(\frac{\hat{v}}{k}\right)k$$

  - $\pi$: profit per product line
  - $\hat{v}$: vacancy to search inventors
  - $c(\cdot)$: increasing and convex

# $dk =$ Internal R&D $+$ Knowledge Diffusion

1. Internal R&D ($-$ Destruction)

$$dk|_{\text{R\&D}} = \left( \underbrace{\mu(n/k)\,k}_{\text{Internal R\&D}} - \underbrace{\delta k}_{\text{Destruction}} \right) dt + \underbrace{\sigma k\, dW}_{\text{Shocks}}$$

   - $\mu(\cdot)$: non-negative, increasing, and concave
   - $\delta$: determined in GE so that the total mass of product lines stays at one
   - Reminiscent of Klette and Kortum (2004)
   - $\underbrace{\mu(n/k)\,k/n}_{\text{R\&D outcome per inventor}}$ is increasing in $k/n \to k/n$: inventor productivity

2. Knowledge diffusion through inventor moves
   - When inventors move from a $(k', n')$-firm to a $(k, n)$-firm, the hiring firm gains $\alpha(k'/n')$ units of knowledge capital per inventor: $dk/dn|_{(k',n')\to(k,n)} = \alpha(k'/n')$
   - The source firm keeps its knowledge capital

# Matching Technology

- Random search
- Only on-the-job search
- Labor market tightness $\theta = (\text{Total mass of vacancies } V) / (\text{Total mass of inventors } N)$
- A vacancy meets an inventor at rate $q(\theta)$
- An inventor meets a vacancy at rate $\theta q(\theta)$

# Bargaining and Non-Compete Enforcement

**Assumption.**

1. In a meeting, the match is formed if the match surplus is positive
2. Poaching firm receives share $(1-\beta)$ of the match surplus, and the poached firm and its inventor receive share $\beta$
3. Vacancy posting decision maximizes the joint value of firm and incumbent inventors

- Without an enforceable non-compete clause, Bertrand competition gives $\beta = 0$ (Postel-Vinay and Robin, 2002)
- A non-compete clause secures $\beta > 0$ because the poaching firm must buy out the clause (Shi, 2023)
- Assumptions follow Lentz and Mortensen (2022); Bilal et al. (2022) microfound assumption 3
- Decisions maximize the joint value of the firm and its inventors, so the firm's state is $(k, n)$, not the distribution of wage contracts

# HJB Equation

$\Omega(k,n,t)$: joint value of $n$ inventors and firm with knowledge capital $k$

$$r(t)\Omega(k,n,t) - \frac{\partial}{\partial t}\Omega(k,n,t) =$$

$$\max_{\hat{v}\geq 0} \underbrace{\pi k - c\left(\frac{\hat{v}}{k}\right)k}_{\text{Profit}-\text{Search cost}} + \underbrace{\{\mu(n/k) - \delta\}k\Omega_k(k,n,t) + \frac{\sigma^2}{2}k^2\Omega_{kk}(k,n,t)}_{\text{Internal R\&D}-\text{Destruction}}$$

$$+ \underbrace{(1-\beta)q(\theta)\hat{v}\int \hat{M}(k,n,k',n',t)dG_n(k',n',t)}_{\text{Poaching}} + \underbrace{\beta\theta q(\theta)n\int \hat{M}(k',n',k,n,t)dG_v(k',n',t)}_{\text{Being poached}}$$

$$\underbrace{\hat{M}(k,n,k',n',t)}_{\text{Match surplus}} = \Big[\underbrace{\Omega_n(k,n,t)}_{\substack{\text{destination}\\ \text{inventor value}}} + \underbrace{\alpha(k'/n')\Omega_k(k,n,t)}_{\text{value of the knowledge brought}} - \underbrace{\Omega_n(k',n',t)}_{\substack{\text{source}\\ \text{inventor value}}}\Big]^+$$

- $G_n(k',n',t)$: inventor-weighted firm distribution
- $G_v(k',n',t)$: vacancy-weighted firm distribution

# A continuum of product lines makes $x = n/k$ a sufficient state variable

- $x \equiv n/k$ is inventors per product line; $x^{-1} = k/n$ is inventor productivity
- With a continuum of product lines and proportional shocks, $d \log k$ and $d \log n$ depend only on $x$, and $\Omega(k, n) = k\, S(x)$
- Firms never lose all product lines, so the model abstracts from entry, exit, and unemployment
- Klette and Kortum (2004): with discrete product lines, the variance of firm growth falls with size, so both $k$ and $n$ are state variables

# Simplified HJB Equation

In the steady state, the joint value per product line $S(x)$ satisfies

$$\rho S(x) = \max_{v \geq 0} \underbrace{\pi - c(v)}_{\text{Profit}-\text{Search cost}} + \underbrace{\{\mu(x) - \delta\}\{S(x) - xS_x(x)\} + \frac{\sigma^2}{2}x^2 S_{xx}(x)}_{\text{Internal R\&D}-\text{Destruction}}$$

$$+ \underbrace{(1-\beta)q(\theta)v\int M(x,x')dF_x(x')}_{\text{Poaching}} + \underbrace{\beta\theta q(\theta)x\int M(x',x)dF_v(x')}_{\text{Being poached}}$$

$$\underbrace{M(x,x')}_{\text{Match surplus}} = \left[\ S_x(x) + \alpha(x')\{S(x) - xS_x(x)\} - S_x(x')\ \right]^+$$

- $v$: vacancy per product line; $F(x)$: distribution of product lines over $x$
- $F_x(x)$, $F_v(x)$: inventor-weighted and vacancy-weighted product line distributions

# Knowledge makes downward moves profitable

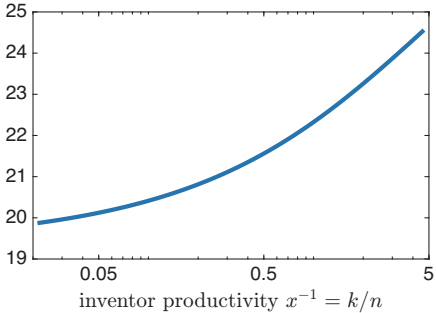
- A move from $x'$ to $x$ has match surplus

$$M(x,x') = \Big[ \underbrace{S_x(x)}_{\substack{\text{destination}\\ \text{inventor value}}} + \underbrace{\alpha(x')\{S(x) - xS_x(x)\}}_{\text{value of the knowledge brought}} - \underbrace{S_x(x')}_{\substack{\text{source}\\ \text{inventor value}}} \Big]^+$$

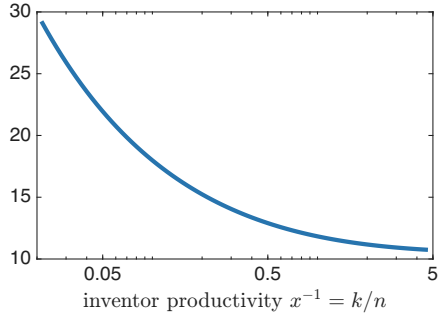
- $S_x(x) = \Omega_n$ and $S(x) - xS_x(x) = \Omega_k$
- If $\alpha = 0$, inventors move only to firms with higher inventor value $S_x$, as in a job ladder
- With $\alpha > 0$, a move to a firm with lower $x^{-1} = k/n$ occurs when the value of the knowledge brought exceeds the loss in inventor value

# Less productive firms value inventors less but knowledge more

Inventor value $S_x(x) = \Omega_n$




Knowledge value $S(x) - xS_x(x) = \Omega_k$




Marginal values by $\beta$

# Mobility shapes growth through reallocation and diffusion

- A Cobb–Douglas aggregator and a quality ladder with step $\lambda$ give growth

$$g = \underbrace{\log(\lambda)}_{\text{innovation step}} \times \underbrace{(\delta_{RD} + \delta_{KD})}_{\text{creative destruction rate}}$$

$$\underbrace{\delta_{RD}}_{\text{internal R\&D}} = \int \mu(x)\, dF(x), \qquad \underbrace{\delta_{KD}}_{\text{knowledge diffusion}} = N\theta q(\theta) \iint \mathbb{1}_P(x, x')\, \alpha(x')\, dF_x(x')\, dF_v(x)$$

- $\mathbb{1}_P(x, x')$: equals 1 when the match surplus $M(x, x')$ is positive
- Since $\mu$ is concave, $\delta_{RD}$ is highest when inventor productivity is equal across product lines

# Sources of Private-Social Wedges

| Private-social wedge | Source | Effect on private incentives |
|---|---|---|
| Innovation value | Appropriability | Too few vacancies |
| Innovation value | Business stealing | Too many vacancies |
| Vacancy posting | Hold-up from surplus sharing | Too few vacancies |
| Vacancy posting | Search congestion | Too many vacancies |

# Estimation

# What INV-BIO Observes

- **Inventors.** Establishment and dates of each job, daily wage, occupation, education, and age
- **Patents.** Filing date, forward citation counts, co-inventors, and technology classes
- **Establishments.** Employment, mean wage, industry, and region

# Functional Forms

| Function | Description |
|---|---|
| $c(v) = \frac{\bar{c}}{\phi+1} v^{\phi+1}$ | Vacancy cost function |
| $q(\theta) = \bar{q}\theta^{\eta-1}$ | Matching function |
| $\mu(x) = \bar{\mu} x^{1/2}$ | Internal R&D function |
| $\alpha(x') = \bar{\alpha} x'^{-1}$ | Knowledge diffusion function |

# From the Law of Motion to a Regression

- $e$: establishment; $t$: year
- Inventor productivity $z \equiv x^{-1} = k/n$ is measured by $z_{e,t}$, three-year citations per inventor at $e$ to their patents filed in $t$
- $\alpha(x') = \overline{\alpha}\, z'$ and $\mu(x) = \overline{\mu}\, z^{-1/2}$
- $z^{\text{inc}}_{e,t}$ is the productivity of inventors who stay at $e$ from $t-1$ to $t$
- $m_{e' \to e,t}$ inventors move from $e'$ to $e$ between $t-1$ and $t$

$$\Delta \log z^{\text{inc}}_{e,t} = \underbrace{\overline{\alpha} \sum_{e'} \frac{m_{e' \to e,t}}{n_{e,t-1}} \frac{z_{e',t-1}}{z_{e,t-1}}}_{\text{diffusion exposure}} + 12\,\overline{\mu} \underbrace{z_{e,t-1}^{-1/2}}_{\substack{\text{R\&D} \\ \text{exposure}}} + \underbrace{\text{net inflow rate} + \text{FE}}_{\text{controls}} + \varepsilon_{e,t}$$

# Estimates of Knowledge Diffusion and Internal R&D

| | Incumbent productivity growth | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Diffusion exposure | 0.526*** | 0.306*** | 0.250*** |
| | (0.071) | (0.073) | (0.089) |
| R&D exposure $z_{e,t-1}^{-1/2}$ | 0.255*** | 0.347*** | 0.440*** |
| | (0.023) | (0.024) | (0.029) |
| Establishment FE | Yes | Yes | Yes |
| Year FE | Yes | No | No |
| Industry × region × year FE | No | Yes | Yes |
| Establishment linear trend | No | No | Yes |
| Root MSE | 0.702 | 0.641 | 0.653 |
| Establishment-years | 12,690 | 10,812 | 10,812 |

- Column (1), the model's moment net of establishment and year effects, sets

$$\overline{\alpha} = 0.526$$
$$\overline{\mu} = 0.2546/12 = 0.0212$$
$$\sigma = 0.702/\sqrt{12} = 0.203$$

INV-BIO establishment-years with positive exposure, weighted by incumbent inventors; net inflow rate in all columns; standard errors clustered by establishment. Annual data, monthly model.

Robustness table Source vs. mover

# Incumbent productivity rises with predicted source-contraction inflows

- Predicted moves allocate inventors at severely contracting sources in proportion to each source's moves two to six years earlier

$$\hat{m}_{e' \to e,t} = \underbrace{s_{e' \to e,t}}_{\substack{\text{share of } e'\text{'s} \\ \text{past moves to } e}} \times \underbrace{n_{e',t-1} \, \mathbb{1}\{\text{employment at } e' \text{ falls by at least 80\%}\}}_{\text{displaced inventors}}$$

- Predicted exposure is the diffusion exposure with $\hat{m}_{e' \to e,t}$ in place of $m_{e' \to e,t}$

| Coefficient on predicted exposure | (1) | (2) | (3) |
|---|---|---|---|
| *First stage* | | | |
| Realized diffusion exposure | 0.0236** | 0.0245* | 0.0363 |
| | (0.0117) | (0.0125) | (0.0231) |
| *Reduced form* | | | |
| Incumbent productivity growth | 0.0515** | 0.0482** | 0.0449 |
| | (0.0200) | (0.0187) | (0.0295) |

INV-BIO establishment-years; fixed effects as in baseline columns (1)–(3); standard errors clustered by establishment.

# Calibration

| *Set externally or normalized* | | |
|---|---|---|
| $\rho$ | 0.004 | Monthly discount rate (5% a year) |
| $\overline{c}$ | 1 | Vacancy-cost scale, normalized |
| $\phi$ | 3.45 | Vacancy-cost curvature, Bilal et al. (2022) |
| $\eta$ | 0.50 | Matching elasticity, Petrongolo and Pissarides (2001) |
| $\varepsilon$ | 0.50 | R&D elasticity, Acemoglu et al. (2018) |
| $\pi$ | 0.23 | Profit per product line (markup 1.3) |
| $N$ | 1.95 | Inventors ($1/N = 0.512$ citations) |
| $\beta$ | 0.50 | Baseline surplus share |

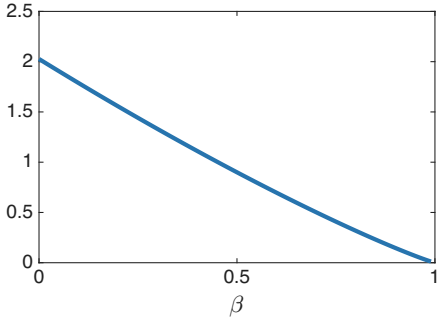
| *Regression, column (1)* | | |
|---|---|---|
| $\overline{\alpha}$ | 0.526 | Knowledge diffusion |
| $\overline{\mu}$ | 0.0212 | Internal R&D |
| $\sigma$ | 0.203 | R&D shocks |
| *Balanced-growth targets* | | |
| $\overline{q}$ | 0.0198 | Job-to-job rate 12.5% (2001) |
| $\lambda$ | 1.053 | Growth 2% a year |

- With $\beta = 0.5$, a ban raises the job-to-job rate by 10.2%, close to Hawaii's 11% rise (Balasubramanian et al., 2022)

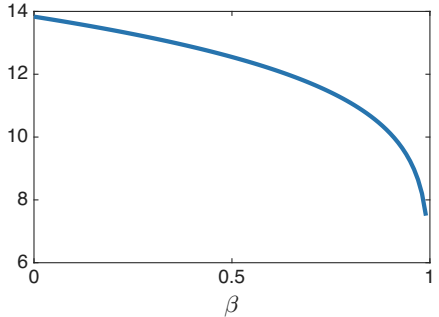
# Quantitative Exercises

# Stronger non-compete enforcement lowers recruiting and mobility

Vacancy costs (% of output)



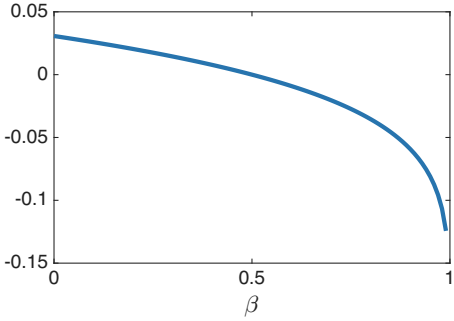

Job-to-job transition rate (%)



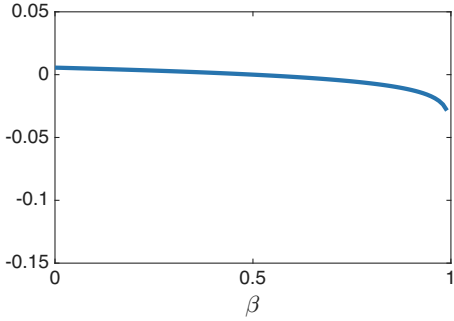

- The poaching firm receives a share $1-\beta$ of the match surplus
- Steady states by $\beta$

# Stronger enforcement lowers growth, mostly through diffusion

Knowledge-diffusion growth, change from $\beta = 0.5$ (pp)



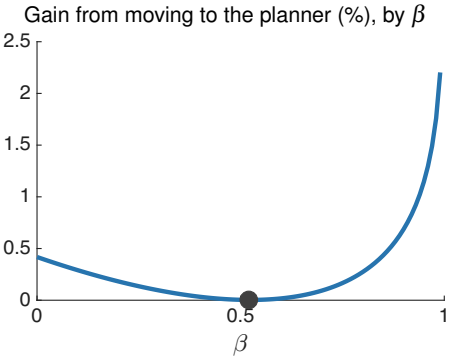

Internal-R&D growth, change from $\beta = 0.5$ (pp)




Steady states, annual rates on a common vertical scale.

- At $\beta = 0.5$, knowledge diffusion accounts for 16% of growth
- However, most of the change in growth comes from knowledge diffusion
- Internal-R&D growth responds only through productivity dispersion, which moves little

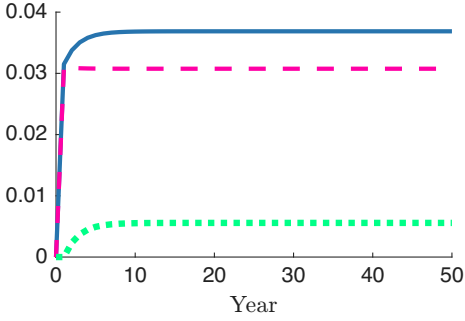
# A fixed split nearly implements the planner



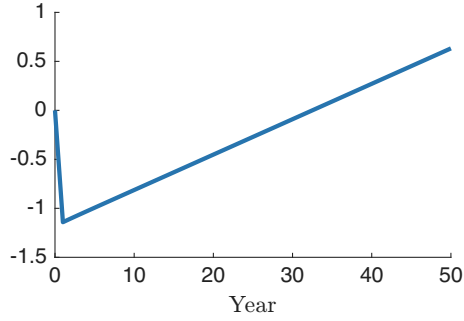

- For each $\beta$, the figure shows the CE gain from moving from its steady state to the planner
- The gain is smallest at $\beta^* = 0.52$, at 0.0029% of consumption
- A ban raises welfare only if $\beta$ exceeds 0.85 before the ban

# A ban raises growth and lowers welfare

Growth (pp): total, diffusion (dashed), R&D (dotted)




Log consumption, change from initial BGP ($\times 100$)

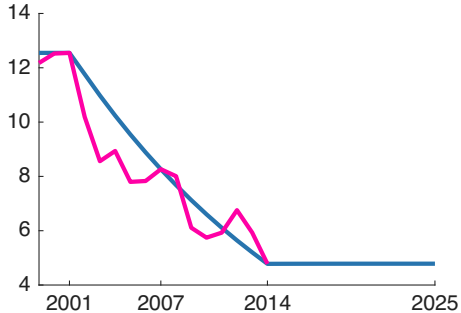



- $\beta$ falls permanently from 0.5 to 0, and the job-to-job rate rises by 10.2%
- Long-run growth rises by 0.037 pp, 0.031 pp of it through knowledge diffusion
- Firms spend more on recruiting, so CE welfare falls by 0.41%

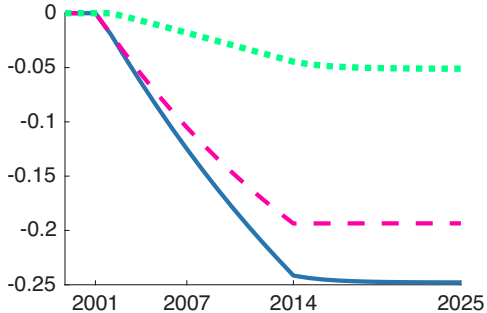
All transition panels Welfare measure

# Declining inventor mobility lowers growth and welfare

Job-to-job rate (%): model (blue), data (magenta)




Growth (pp): total, diffusion (dashed), R&D (dotted)




- A path of matching efficiency matches the rates in 2001, 2007, and 2014 (12.5%, 8.3%, 4.8%)
- Long-run growth falls by 0.25 pp, 0.19 pp of it through knowledge diffusion
- CE welfare falls by 3.46%

Matching-efficiency path Annual mobility rates

# Conclusion

- Inventors often move to lower-wage establishments yet gain more in wages than other movers
- The efficient fixed split of the match surplus nearly implements the planner ($\beta^* = 0.52$)
- A non-compete ban raises growth and lowers welfare by 0.41%
- The 2001–2014 inventor mobility decline lowers growth by 0.25 pp and welfare by 3.46%
- Knowledge diffusion, not reallocation, drives most of both growth effects

# Appendix

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# Summary Statistics: INV-BIO

| Establishment-level variables | Mean | S.D. | $N$, thousand |
|---|---|---|---|
| $N$ of inventors ($n_{et}$) | 5.18 | 29.53 | 678.9 |
| $N$ of employees | 300.76 | 1088.36 | 678.9 |
| Mean daily wage, Euro | 100.83 | 54.63 | 670.3 |
| Three-year backward average citation productivity ($\bar{z}_{et}^{(3)}$) | 0.13 | 0.66 | 465.6 |
| Total inventor inflows | 0.40 | 2.99 | 678.9 |
| Inventor productivity growth | $-0.02$ | 1.09 | 13.9 |
| Knowledge-diffusion inflow term | 0.16 | 0.34 | 13.9 |
| Net inventor inflow rate | 0.06 | 0.14 | 13.9 |

$$z_{et} = \frac{1}{|J_{et}|} \sum_{j \in J_{et}} \sum_{i \in I_{jt}} \frac{\text{three-year citations}_i}{\text{inventors}_i}$$

$J_{et}$: inventors at $e$ in year $t$; $I_{jt}$: patents filed by inventor $j$ in year $t$.

Back to data Back to regression

# Fact 1: Moves with Own-Wage Gains

Ranked by citation productivity (row %)

| | Destination rank bin | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Origin | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 1 | 12 | 11 | 10 | 10 | 8 | 9 | 9 | 10 | 10 | 10 |
| 2 | 11 | 11 | 11 | 9 | 9 | 9 | 9 | 9 | 10 | 10 |
| 3 | 9 | 11 | 11 | 11 | 9 | 11 | 9 | 10 | 10 | 10 |
| 4 | 11 | 10 | 11 | 11 | 10 | 10 | 11 | 9 | 9 | 8 |
| 5 | 8 | 8 | 9 | 8 | 9 | 11 | 16 | 10 | 9 | 10 |
| 6 | 7 | 8 | 10 | 7 | 11 | 10 | 10 | 12 | 11 | 14 |
| 7 | 5 | 7 | 6 | 14 | 12 | 9 | 12 | 11 | 11 | 12 |
| 8 | 5 | 6 | 7 | 6 | 8 | 13 | 12 | 15 | 15 | 14 |
| 9 | 4 | 5 | 11 | 6 | 10 | 8 | 9 | 17 | 12 | 18 |
| 10 | 4 | 7 | 5 | 6 | 5 | 10 | 10 | 11 | 16 | 26 |

Ranked by mean wage (row %)

| | Destination rank bin | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Origin | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 1 | 17 | 16 | 13 | 12 | 10 | 9 | 7 | 7 | 6 | 4 |
| 2 | 10 | 16 | 13 | 12 | 10 | 9 | 8 | 8 | 7 | 6 |
| 3 | 7 | 12 | 13 | 12 | 10 | 9 | 9 | 9 | 9 | 9 |
| 4 | 5 | 9 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 9 |
| 5 | 5 | 8 | 9 | 11 | 10 | 11 | 10 | 12 | 12 | 13 |
| 6 | 3 | 7 | 8 | 9 | 10 | 13 | 13 | 12 | 13 | 12 |
| 7 | 3 | 6 | 7 | 8 | 9 | 11 | 13 | 13 | 15 | 16 |
| 8 | 2 | 5 | 5 | 7 | 10 | 11 | 14 | 14 | 15 | 18 |
| 9 | 2 | 4 | 5 | 5 | 9 | 9 | 14 | 14 | 16 | 22 |
| 10 | 2 | 3 | 4 | 4 | 5 | 7 | 11 | 12 | 18 | 34 |

INV-BIO, 1980–2014, inventor job-to-job moves in which the inventor's own wage rises.

Back to Fact 1

# SIAB Data: Summary Statistics

| Worker-level variables | Mean | S.D. | $N$, thousand |
|---|---|---|---|
| $D_{it}^{\text{wage}}$ | 0.45 | – | 652 |
| $\Delta \log w_{it}$ | 0.173 | 0.657 | 652 |
| Wage increase indicator | 0.72 | – | 652 |
| $I_{i,t-1}$ | 0.10 | – | 652 |
| Age | 33.3 | 11.3 | 652 |
| Female, % | 42.7 | – | 652 |
| University-entrance qualification, % | 27.0 | – | 652 |
| Log wage | 3.877 | 0.871 | 652 |

Back to data

# Definition of Inventor-Proxy Workers

- Inventor-proxy workers: SIAB workers with a university-entrance qualification in the five largest KldB 2010 parent occupation groups among INV-BIO inventors
  - Machine and vehicle technology (25)
  - Mechatronics, energy and electrical engineering (26)
  - Technical R&D, design and production planning (27)
  - Mathematics, biology, chemistry and physics (41)
  - Business management and organisation (71)
- These groups account for 80.5% of inventor-years with observed occupations

# Inventor-Proxy Workers in SIAB and Inventors in INV-BIO

| Summary statistics (1980–2014) | | SIAB | | INV-BIO |
|---|---|---|---|---|
| | | Workers | Inventor-proxy workers | Inventors |
| Wage, Euro | Mean | 63.8 | 99.5 | 123.8 |
| | S.D. | 45.5 | 55.6 | 52.1 |
| Age | Mean | 38.5 | 39.0 | 38.9 |
| | S.D. | 12.6 | 11.2 | 10.6 |
| Females, % | | 44.2 | 33.7 | 6.4 |
| University-entrance qualification, % | | 15.3 | 100.0 | 70.3 |
| *N* of observations, thousand | | 17,474 | 1,030 | 3,517 |

Inventor-proxy workers are workers whose observed KldB 2010 parent occupation group is 25, 26, 27, 41, or 71 and who have a university-entrance qualification.

# Fact 2: Regression Specification

- Compare inventor-proxy workers with other job-to-job movers

$$y_{it} = b\, l_{i,t-1} + X'_{i,t-1}\gamma + \mathrm{FE}_{o(i)} + \mathrm{FE}_{d(i)} + \mathrm{FE}_t + \varepsilon_{it}$$

- $y_{it}$: lower-mean-wage destination or wage growth
- Sample: BeH-to-BeH establishment changes without unemployment spell
- $X_{i,t-1}$: age, age squared, gender, and log wage

Back to Fact 2

# Inventor-Proxy Worker Flows: Wage-Increase Moves

| | Wage-increase job-to-job sample | |
|---|---|---|
| | (1) | (2) |
| Dependent variable | $D_{it}^{\text{wage}}$ | $\Delta \log w_{it}$ |
| $I_{i,t-1}$ | 0.0094*** | 0.0535*** |
| | (0.0025) | (0.0027) |
| Controls | Yes | Yes |
| Origin estab. FE | Yes | Yes |
| Dest. estab. FE | Yes | Yes |
| Year FE | Yes | Yes |
| Observations | 404,722 | 404,722 |
| Adjusted $R^2$ | 0.5678 | 0.7002 |

Sample is restricted to establishment changes with an increase in the worker's own daily wage. Other definitions match the full job-to-job sample.

Back to Fact 2

# Inventor Mass per Product Line $x$ as State Variable

- $x \equiv n/k$: inventor mass per product line
- $x^{-1} = k/n$: inventor productivity
- $f(x)$: distribution of product lines over $x$
- $S(x)$: joint value of $x$ mass of inventors and unit mass of product line

$$\underbrace{kg(k,n)}_{\text{\# of product line} \times \text{Firm distribution}} = \underbrace{f(x)}_{\text{Product distribution}}$$

$$\underbrace{\Omega(k,n)}_{\text{Firm-level joint value}} = \underbrace{kS(x)}_{\text{\# of product line} \times \text{Product-level joint value}}$$

$$\underbrace{\Omega_n(k,n)}_{\text{Marginal value of inventor}} = \underbrace{S_x(x)}_{\text{Inventor per product line} \uparrow}$$

$$\underbrace{\Omega_k(k,n)}_{\text{Marginal value of knowledge capital}} = \underbrace{S(x)}_{\text{Product line} \uparrow} \quad \underbrace{-xS_x(x)}_{\text{Inventor per product line} \downarrow}$$

Back to simplified HJB

# Aggregation Setting

- Representative household utility

$$\int_0^\infty e^{-\rho t} \log C(t) dt$$

- Inelastically supplies $L$ units of production labor and $N$ inventors
- Aggregate output

$$\log Y(t) = \int_0^1 \log \left( a(\omega, t) l(\omega, t) \right) d\omega$$

  - $a(\omega, t)$: productivity of product $\omega$
  - $l(\omega, t)$: production labor used to produce product $\omega$ (no friction)

- Arrival of creative destruction is random across product lines
- Innovation step size is $\lambda > 0$

# BGP Equilibrium 1

*A stationary competitive equilibrium* consists of sequence of joint value, vacancy policy, poaching indicator function, inventor productivity distribution, creative destruction rate, and labor market tightness $\left[\left\{S(x), v(x), \{\mathbb{1}_P(x,x')\}_{x'\in(0,\infty)}, f(x)\right\}_{x\in(0,\infty)}, \delta, \theta\right]$ such that

- The joint value $S(x)$ satisfies the HJB equation

$$\begin{aligned}\rho S(x) =& \pi - c(v(x)) + \{\mu(x) - \delta\}\{S(x) - xS_x(x)\} + \frac{\sigma^2}{2}x^2 S_{xx}(x) \\ &+ (1-\beta)q(\theta)v(x)\int M(x,x')dF_x(x') + \beta\theta q(\theta)x\int M(x',x)dF_v(x')\end{aligned}$$

where the match surplus $M(x,x')$ is given by

$$M(x,x') = \left[S_x(x) + \alpha(x')\{S(x) - xS_x(x)\} - S_x(x')\right]^+$$

# BGP Equilibrium 2

- The vacancy posting is optimal, so that $v(x,t)$ satisfies

$$c'(v(x)) = (1-\beta)\, q(\theta) \int M(x,x')dF_x(x')$$

- Given a match, the inventor moves if the match surplus is positive:

$$\mathbb{1}_P(x,x') = \begin{cases} 1 & \text{if} \quad M(x,x') > 0 \\ 0 & \text{otherwise} \end{cases}$$

# BGP Equilibrium 3

- The inventor productivity distribution $f(x,t)$ follows the KF equation

$$
\begin{aligned}
0 = & \left\{ \begin{array}{c} \mu(x) - \delta \\ +q(\theta) v(x) \int \mathbb{1}_P(x,x') \alpha(x') dF_x(x') \end{array} \right\} f(x) \\
& - \frac{\partial}{\partial x} \left[ \left\{ \begin{array}{c} -\{\mu(x) - \delta\} x \\ +q(\theta) v(x) \int \mathbb{1}_P(x,x') \{1 - \alpha(x') x\} dF_x(x') \\ -\theta q(\theta) x \int \mathbb{1}_P(x',x) dF_v(x') \end{array} \right\} f(x) \right] \\
& + \frac{\partial^2}{\partial x^2} \left[ \frac{\sigma^2}{2} x^2 f(x) \right],
\end{aligned}
$$

- The labor market tightness $\theta$ and the creative destruction rate $\delta$ satisfy

$$
\begin{aligned}
N\theta &= \int v(x) f(x) dx \\
\delta &= \int \mu(x) dF(x) + N\theta q(\theta) \int \int \mathbb{1}_P(x,x') \alpha(x') dF_x(x') dF_v(x)
\end{aligned}
$$

# Social Optimum

$$\rho S^*(x) = \max_{v \geq 0} \underbrace{\pi^*}_{\text{Growth externality}} - c(v(x)) + \{\mu(x) - \delta\}\{S^*(x) - xS_x^*(x)\} + \frac{\sigma^2}{2}x^2 S_{xx}^*(x)$$

$$+ \underbrace{\theta q'(\theta) v(x) \int\int M^*(x, x')\ dF_x(x') dF_v(x)}_{\text{Congestion externality}}$$

$$+ \underbrace{q(\theta) v(x) \int M^*(x, x')\, dF_x(x') + \theta q(\theta) x \int M^*(x', x)\, dF_v(x')}_{\text{Hold-up problem}}$$

- Externality from Schumpeterian growth: appropriability effect + business stealing effect
- Externality from random search: congestion externality + hold-up problem

# Social Planner's Problem

$$\mathbf{x}(t) = \left[\log Y(t), \{f(x,t)\}_{x\in(0,\infty)}\right]$$

$$\mathbf{y}(t) = \left[\delta(t), \theta(t), \{v(x,t)\}_{x\in(0,\infty)}, \left\{\mathbb{1}_P(x,x',t)\right\}_{x,x'\in(0,\infty)}\right]$$

$$\boldsymbol{\lambda}_{\mathbf{x}}(t) = \left[\lambda_Y(t), \{\lambda_f(x,t)\}_{x\in(0,\infty)}\right]$$

$$\boldsymbol{\lambda}_{\mathbf{y}}(t) = [\lambda_\delta(t), \lambda_\theta(t)]$$

Back to social optimum

# Social Planner's Problem

$$
\begin{aligned}
&\mathscr{H}^{P}\left(\mathbf{x}(t),\mathbf{y}(t),\boldsymbol{\lambda}_{\mathbf{x}}(t),\boldsymbol{\lambda}_{\mathbf{y}}(t)\right)\\
&=\ln Y(t)+\ln\left\{1-\int c\left(v(x,t)\right)dF(x,t)\right\}\\
&\quad+\lambda_{Y}(t)\,\delta(t)\ln\lambda\\
&\quad+\lambda_{\delta}(t)\left[-\delta(t)+\int\left\{\mu(x)+q(\theta(t))v(x,t)\int\mathbb{1}_{P}(x,x',t)\alpha\left(x'\right)dF_{x}(x',t)\right\}dF(x,t)\right]\\
&\quad+\lambda_{\theta}(t)\left\{N\theta(t)-\int_{0}^{\infty}v(x,t)\,f(x,t)dx\right\}\\
&\quad+\int_{0}^{\infty}\lambda_{f}(x,t)\left[\begin{array}{c}\left\{\begin{array}{c}q(\theta(t))v(x,t)\int\mathbb{1}_{P}(x,x',t)\alpha\left(x'\right)dF_{x}(x',t)\\+\mu(x)-\delta(t)\end{array}\right\}f(x,t)\\-\frac{\partial}{\partial x}\left[\left\{\begin{array}{c}q(\theta(t))v(x,t)\int\mathbb{1}_{P}(x,x',t)\left\{1-\alpha\left(x'\right)x\right\}dF_{x}(x',t)\\-\theta(t)\,q(\theta(t))x\int\mathbb{1}_{P}(x',x,t)dF_{v}(x',t)\\-\left\{\mu(x)-\delta(t)\right\}x\end{array}\right\}f(x,t)\right]\\+\frac{\partial^{2}}{\partial x^{2}}\left[\frac{\sigma^{2}}{2}x^{2}f(x,t)\right]\end{array}\right]dx
\end{aligned}
$$

Back to social optimum

# Social Planner's Problem

$$
\begin{aligned}
0 &= \frac{\partial}{\partial \mathbf{y}(t)} \mathscr{H}^P\left(\mathbf{x}(t), \mathbf{y}(t), \boldsymbol{\lambda}_{\mathbf{x}}(t), \boldsymbol{\lambda}_{\mathbf{y}}(t)\right) \\
\rho \boldsymbol{\lambda}_{\mathbf{x}}(t) - \dot{\boldsymbol{\lambda}}_{\mathbf{x}}(t) &= \frac{\partial}{\partial \mathbf{x}(t)} \mathscr{H}^P\left(\mathbf{x}(t), \mathbf{y}(t), \boldsymbol{\lambda}_{\mathbf{x}}(t), \boldsymbol{\lambda}_{\mathbf{y}}(t)\right) \\
0 &= \frac{\partial}{\partial \boldsymbol{\lambda}_{\mathbf{y}}(t)} \mathscr{H}^P\left(\mathbf{x}(t), \mathbf{y}(t), \boldsymbol{\lambda}_{\mathbf{x}}(t), \boldsymbol{\lambda}_{\mathbf{y}}(t)\right) \\
\dot{\mathbf{x}}(t) &= \frac{\partial}{\partial \boldsymbol{\lambda}_{\mathbf{x}}(t)} \mathscr{H}^P\left(\mathbf{x}(t), \mathbf{y}(t), \boldsymbol{\lambda}_{\mathbf{x}}(t), \boldsymbol{\lambda}_{\mathbf{y}}(t)\right) \\
0 &= \lim_{t \to \infty} \left[\exp\left(-\rho t\right) \boldsymbol{\lambda}_{\mathbf{x}}(t) \mathbf{x}(t)\right]
\end{aligned}
$$

Back to social optimum

# Optimal Instantaneous Profit

$$\underbrace{\frac{\ln \lambda}{\rho}\left[1-\int_0^{\infty} c(v(x))\, f(x) dx\right]}_{\text{marginal gain from innovation}} = \underbrace{\int_0^{\infty}\left[S^*(x)-S_x^*(x)\, x\right] f(x) dx}_{\text{expected marginal value of knowledge}}$$

Back to social optimum

# Decomposition of Firm Growth

$$d\log x^{-1}(x,t) = d\log k(x,t) - d\log n(x,t)$$

$$d\log k(x,t) = \underbrace{q(\theta(t))v(x,t)\int \mathbb{1}_P(x,x',t)\alpha(x')\,dF_x(x',t)dt}_{\text{Knowledge Gain from Poaching}} + \underbrace{\left(\mu(x) - \delta(t) - \frac{\sigma^2}{2}\right)dt + \sigma dW(t)}_{\text{R\&D}-\text{Creative Destruction}}$$

$$d\log n(x,t) = \underbrace{\frac{q(\theta(t))v(x,t)}{x}\int \mathbb{1}_P(x,x',t)dF_x(x',t)dt}_{\text{inventor inflow rate}} - \underbrace{\theta(t)\,q(\theta(t))\int \mathbb{1}_P(x',x,t)dF_v(x',t)dt}_{\text{inventor outflow rate}}$$

Back to the regression

# The source's productivity, not the mover's, drives the estimate

| | Incumbent productivity growth | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Source productivity, leave-one-mover-out | 0.478*** | 0.273*** | 0.210** |
| | (0.070) | (0.080) | (0.099) |
| Mover's own productivity | 0.048 | 0.040 | 0.036 |
| | (0.030) | (0.039) | (0.049) |
| Establishment-years | 12,628 | 10,757 | 10,757 |

INV-BIO establishment-years; fixed effects as in baseline columns (1)–(3); standard errors clustered by establishment.

- Without the mover, the source coefficient stays near the baseline estimate, while the mover's own coefficient is small

Back to estimate

# Robustness of the Baseline Regression

| | Zero-exposure sample | | | Lagged-outcome control | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| Diffusion exposure | $0.3151^{***}$ | $0.1803^{***}$ | $0.1213^{**}$ | $0.3127^{***}$ | $0.2341^{**}$ |
| | (0.0519) | (0.0514) | (0.0613) | (0.0811) | (0.1006) |
| Lagged incumbent productivity growth | | | | $-0.1982^{***}$ | $-0.1896^{***}$ |
| | | | | (0.0180) | (0.0218) |
| Establishment FE | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | No | No | No | No |
| Industry $\times$ region $\times$ year FE | No | Yes | Yes | Yes | Yes |
| Establishment-specific linear trend | No | No | Yes | No | Yes |
| Establishment-years | 32,588 | 29,907 | 29,907 | 8,860 | 8,860 |

Columns (1)–(3) add establishment-years with zero exposure. All columns include the R&D exposure and the net inflow rate.

Back to estimate

# Untargeted Productivity Dispersion

| Moment | Data | Model |
|---|---|---|
| SD of log positive inventor productivity | 1.08 | 0.89 |
| P90/P50 of positive inventor productivity | 2.94 | 3.24 |
| P50/P10 of positive inventor productivity | 4.81 | 3.15 |

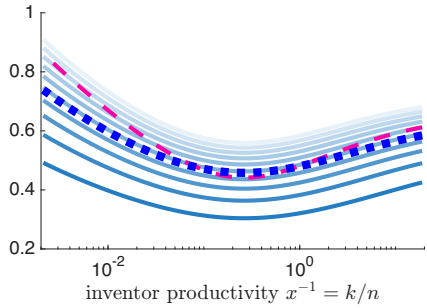
- The model understates overall and lower-tail dispersion and overstates the upper-tail ratio

Data: establishment-years with positive three-year citation productivity, 2001–2011, weighted by inventors. Model: inventor-weighted stationary distribution of $k/n$.

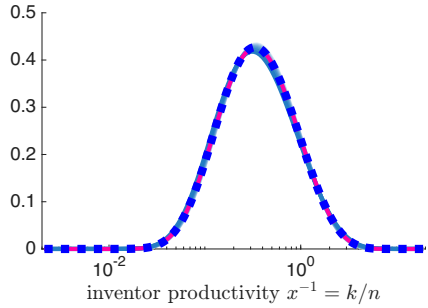
Back to calibration

# Vacancy Posting Policy and Distribution

Vacancy posting policy $v(x)$




Inventor-weighted distribution $f_x(x)$

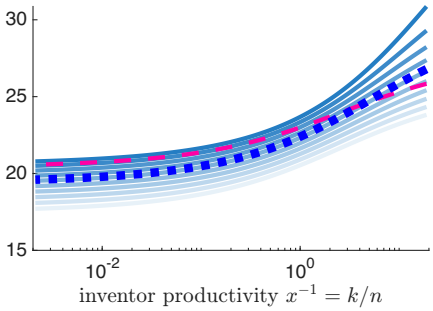



- $\beta \uparrow$ (The share of match surplus of poaching firms $\downarrow$) $\rightarrow$ Vacancy posting $\downarrow$

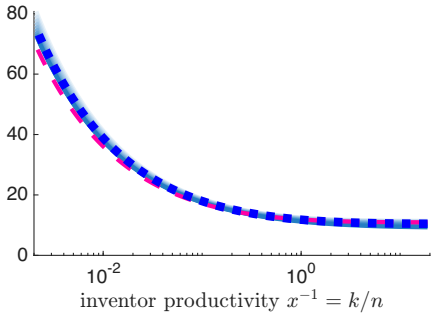
Horizontal axis: inventor productivity $x^{-1} = k/n$. Darker curves: larger $\beta = 0, 0.1, \ldots, 0.9$. Dashed: planner. Dotted: efficient split $\beta^* = 0.52$.

# Marginal Value of Inventor & Marginal Value of Knowledge

$\Omega_n(k,n) = S_x(x)$



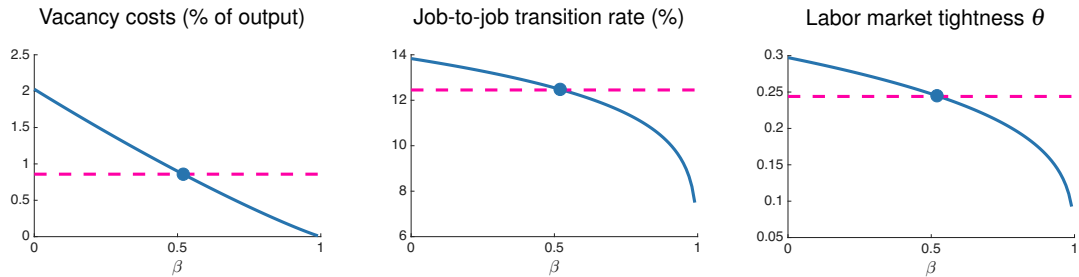

$\Omega_k(k,n) = S(x) - xS_x(x)$




Horizontal axis: inventor productivity $x^{-1} = k/n$. Darker curves: larger $\beta = 0, 0.1, \ldots, 0.9$. Dashed: planner. Dotted: efficient split $\beta^* = 0.52$.

Back to policy functions

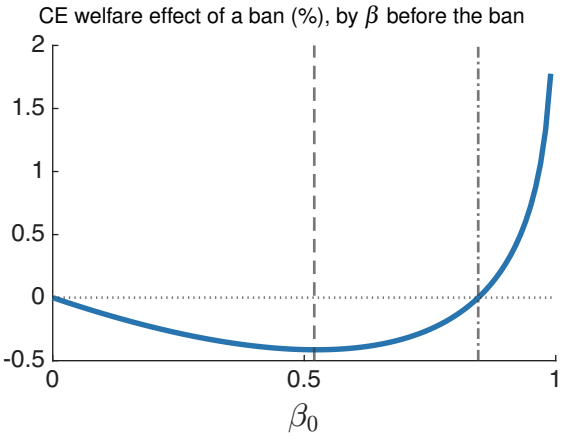
# Vacancy Costs, Job-to-Job Rate, and Tightness




Steady states by $\beta$. Solid: equilibrium; dashed: planner; dot: efficient split $\beta^* = 0.52$.

Back to steady states

# Welfare Effect of a Ban by Pre-Ban Surplus Share




- Every ban leads to the same BGP, so the welfare change tracks pre-ban welfare
- The loss is largest near the efficient split $\beta^* = 0.52$ (dashed line)
- A ban raises welfare only when $\beta$ exceeds 0.85 before the ban (dash-dotted line)

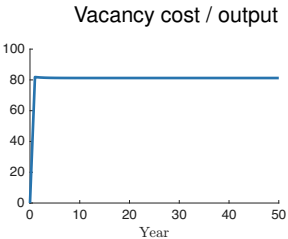
Back to fixed split

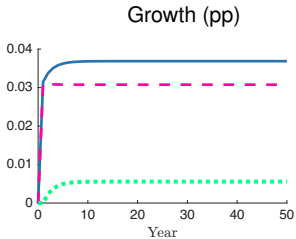
# Efficient Split and Ban Welfare by Matching Elasticity

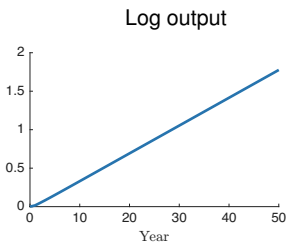
| $\eta$ | $1-\eta$ | $\beta^*$ | Gain at $\beta^*$ (%) | Welfare change from ban (%) | Threshold $\beta_0$ |
|---|---|---|---|---|---|
| 0.3 | 0.70 | 0.71 | 0.0136 | -0.66 | 0.98 |
| 0.4 | 0.60 | 0.62 | 0.0065 | -0.54 | 0.93 |
| 0.5 | 0.50 | 0.52 | 0.0029 | -0.41 | 0.85 |
| 0.6 | 0.40 | 0.43 | 0.0010 | -0.28 | 0.74 |
| 0.7 | 0.30 | 0.34 | 0.0002 | -0.13 | 0.62 |

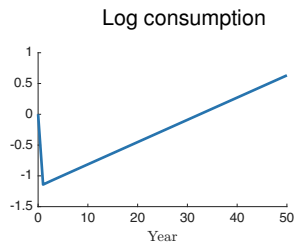
- $\overline{q}$ and $\lambda$ are recalibrated for each $\eta$. $\beta^*$ stays within 0.04 of $1-\eta$, and the ban lowers welfare at $\beta = 0.5$ for every $\eta$

# Transition after a Ban




- Vacancy costs jump on impact; growth rises mainly through knowledge diffusion
- Output rises gradually; consumption falls on impact before recovering
- The growth gain from a ban rises from 0.037 pp at $\beta_0 = 0.5$ to 0.136 pp at $\beta_0 = 0.95$

Levels: 100 times the log deviation from the initial BGP. Growth: solid, total; dashed, knowledge diffusion; dotted, internal R&D. Horizontal axis: years.

Back to the ban

# Consumption-Equivalent Welfare

$$\log C(t) = \log\lambda \int_0^t \delta(\tau)\, d\tau + \log L + \log\Big\{1 - \int c\big(v(x,t)\big)\, dF(x,t)\Big\}$$

$$V = \int_0^\infty e^{-\rho t} \log C(t)\, dt, \qquad \mathscr{L} = \exp\big[\rho\,\big(V(\widetilde{C}) - V(C)\big)\big]$$

- $C$: consumption on the initial BGP; $\widetilde{C}$: consumption after the change. Reported welfare changes are $100(\mathscr{L} - 1)$
- After the simulation horizon, consumption grows at the new steady-state rate

# Matching-Efficiency Path

- The path's form follows Akcigit and Ates (2023)

$$\overline{q}_t = \overline{q}_0 + \frac{\exp(-\nu t/T) - 1}{\exp(-\nu) - 1}\left(\overline{q}_T - \overline{q}_0\right), \qquad t \in [0, T]$$

- $t = 0$ is 2001 and $t = T$ is 2014
- $\overline{q}_0 = 0.0198$ from the baseline; $\nu = 0.588$ and $\overline{q}_T = 0.00858$ match 8.3% in 2007 and 4.8% in 2014
- Unanticipated before $t = 0$, perfect foresight after. Matching efficiency stays at $\overline{q}_T$ after 2014 while the economy converges

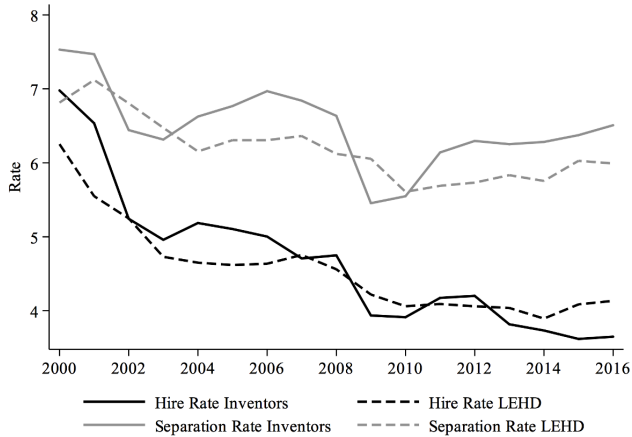
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# German Inventor Job-to-Job Rates

| Year | Inventor person-years | Job-to-job moves | Rate (%) |
|---|---|---|---|
| 1999 | 117,904 | 14,358 | 12.2 |
| 2000 | 126,076 | 15,790 | 12.5 |
| 2001 | 129,050 | 16,194 | 12.5 |
| 2002 | 130,510 | 13,301 | 10.2 |
| 2003 | 130,597 | 11,180 | 8.6 |
| 2004 | 129,933 | 11,611 | 8.9 |
| 2005 | 129,425 | 10,093 | 7.8 |
| 2006 | 129,400 | 10,133 | 7.8 |
| 2007 | 129,446 | 10,699 | 8.3 |
| 2008 | 129,127 | 10,340 | 8.0 |
| 2009 | 127,244 | 7,773 | 6.1 |
| 2010 | 124,402 | 7,146 | 5.7 |
| 2011 | 122,456 | 7,260 | 5.9 |
| 2012 | 120,339 | 8,130 | 6.8 |
| 2013 | 118,021 | 6,998 | 5.9 |
| 2014 | 115,710 | 5,531 | 4.8 |
| Total | 2,009,640 | 166,537 | 8.3 |

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# U.S. Inventor Hire and Separation Rates




Akcigit and Goldschlag (2025), based on the Inventor Employment History, with a 1% LEHD sample for comparison.

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